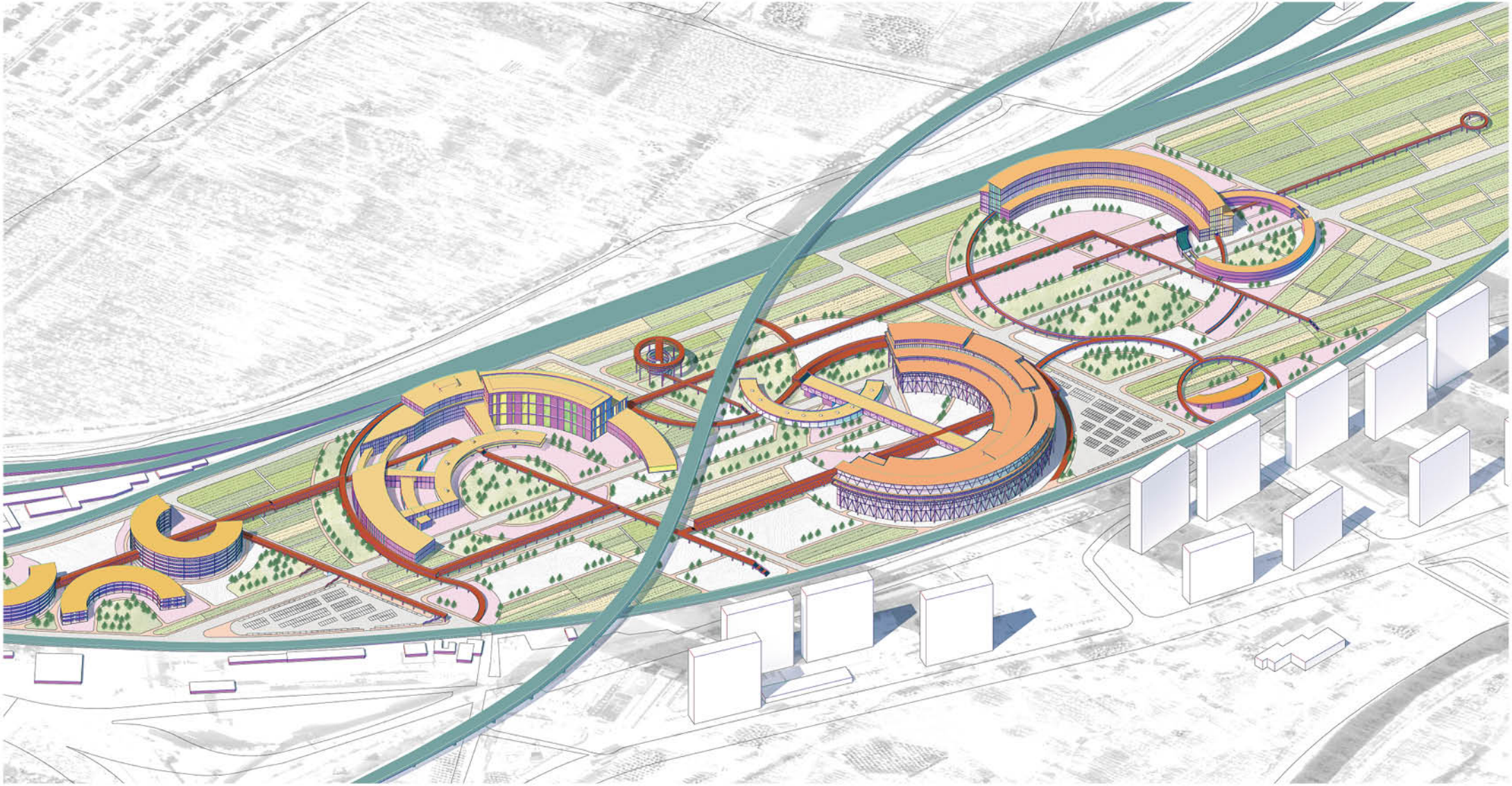


Railway Artists' Self-sufficient Logistics Hub : A Vision For Urban Megastructure

Individual Work

Situated within the 798 relocation area, the Huantie site is characterized by its enclosure by railway tracks, which results in limited connectivity to the surrounding urban fabric. This project seeks to address the site's isolation by proposing the development of a self-sufficient community for young artists, conceptualized within the framework of an urban megastructure. The design integrates multiple functional components, including logistics facilities, catering services, agricultural spaces, and a research center.By breaking the physical and social barriers of the site, the project aspires to transform Huantie into a dynamic and interconnected hub for artistic innovation and sustainable living.

Site Analysis



Three well-developed residential neighourhoods segeragate the site from surounding urban context.



"Huantie Art City" was colsed down.



The inner railway loop serves as a relatively underutilized no-man's land situated between three communes.

Caochangdi International Art Village

Huantie Art City

798 Art District

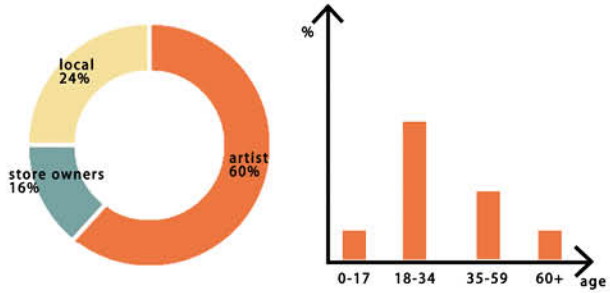
Artistic Neighborhoods Reaching Out.

The entire inner circle of the circular railroad belongs to the junction of three communes and is unmanaged.

Data Reference: Beijing Municipal Commission of Planning and Natural Resources
Beijing Municipal Civil Affairs Bureau
Drafted By Lu

Resource Refinings

1.Flow



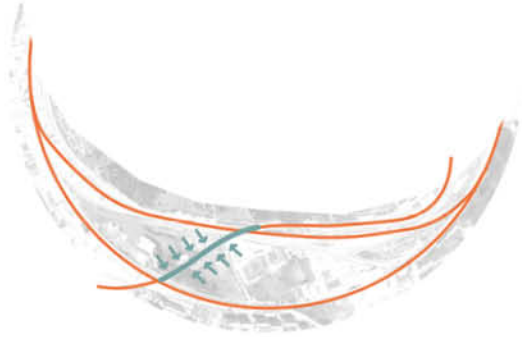
The crowd around the venue is mainly young artists.

2.Farm Land



A significant expanse of rare urban farmland is located in close proximity to the site. Mainly planting cereals and color crops.

3.Railway Heritage

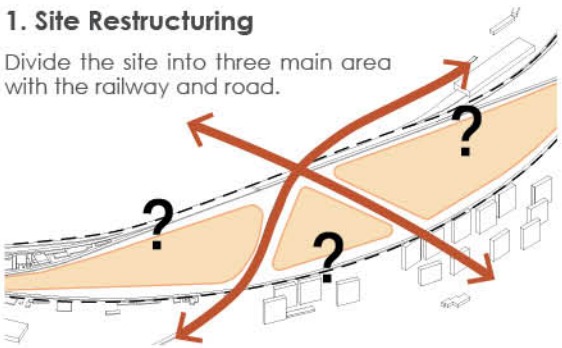


The elevated railroad crossing the site provides a unique landscape for the site.

Form Generation

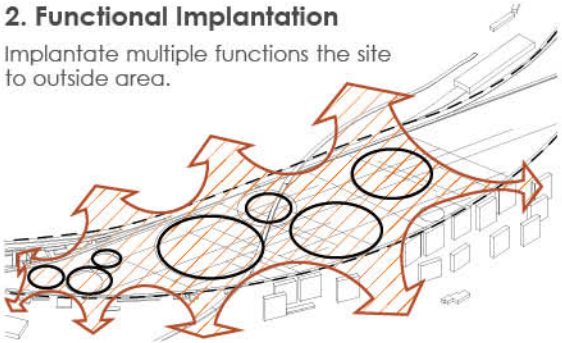
1. Site Restructuring

Divide the site into three main area with the railway and road.



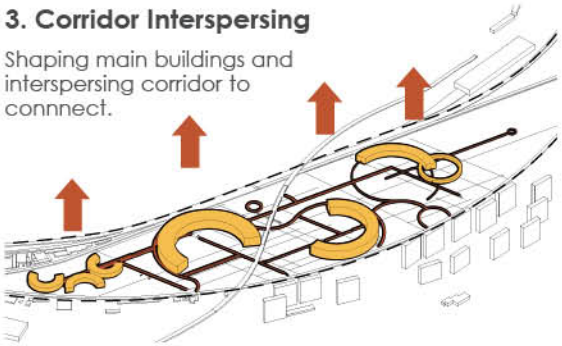
2. Functional Implantation

Implantate multiple functions the site to outside area.



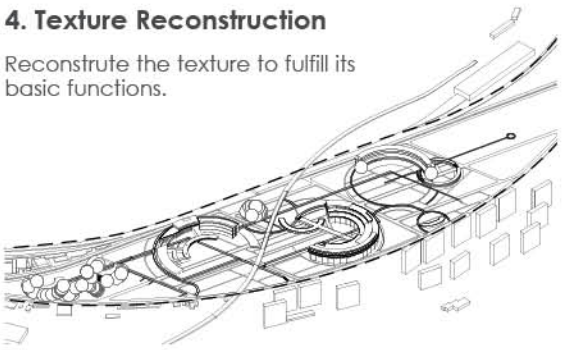
3. Corridor Interspersing

Shaping main buildings and interspersing corridor to connnect.

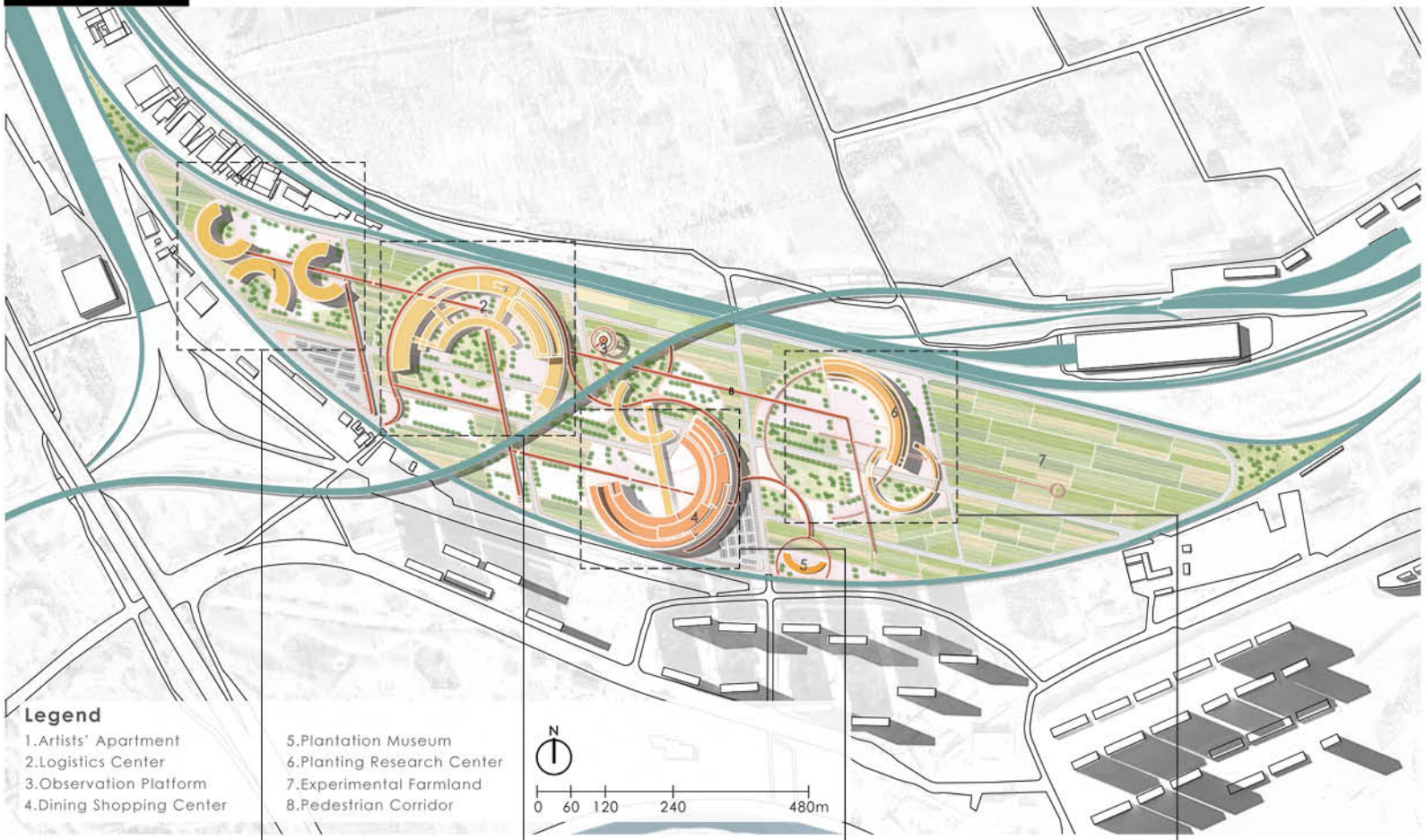


4. Texture Reconstruction

Reconstrute the texture to fulfill its basic functions.



Master Plan





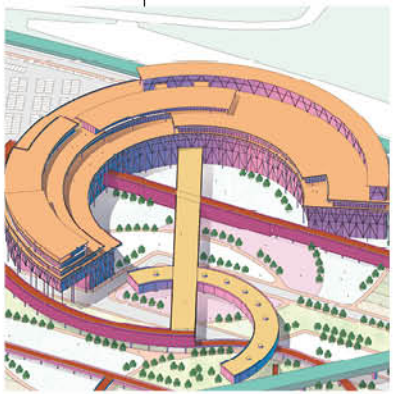
1. Artists' Apartment

Rental apartment for young artists.



2. Logistics Center

Sorting, packaging, office, exhibition and other functions.



3. Shopping Center

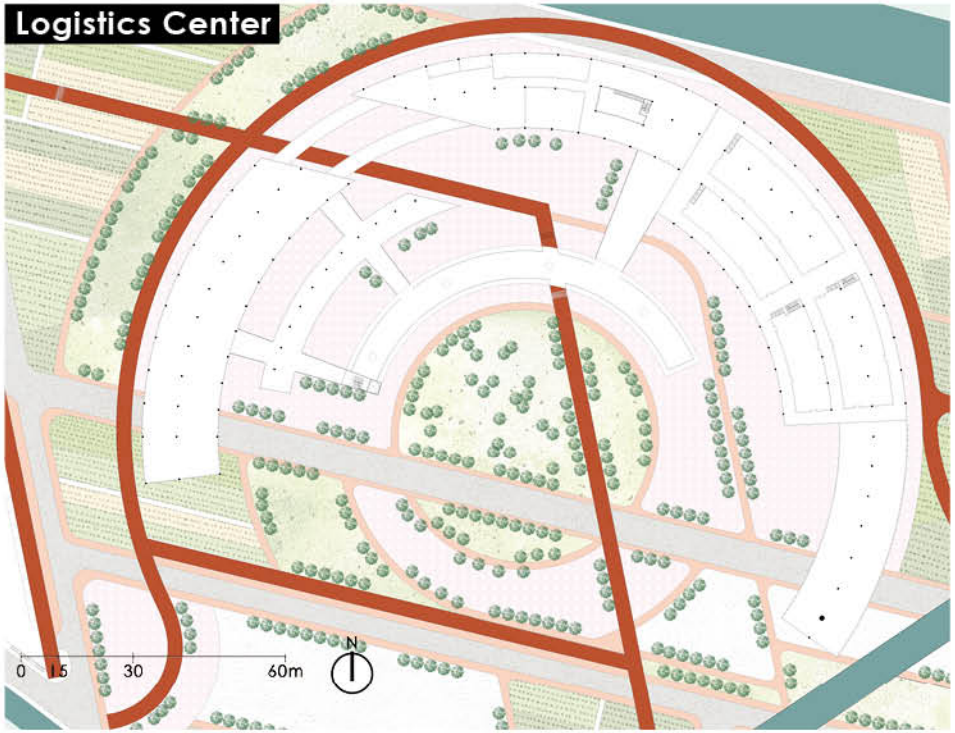
Artists' works and other art supplies for sale, with catering.

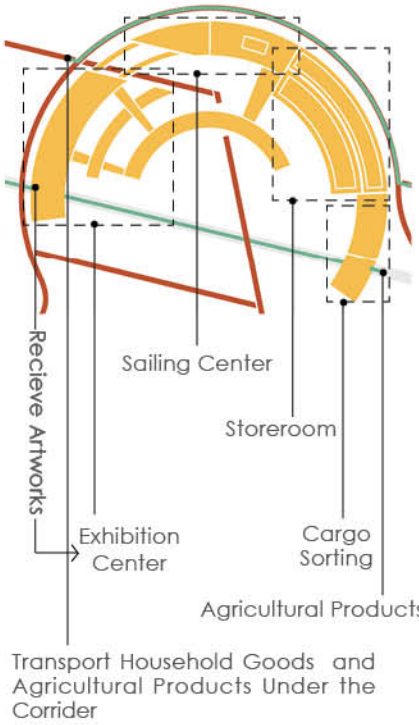


4. Research Center

Growing greenhouses and research centers for pigment crop trials.

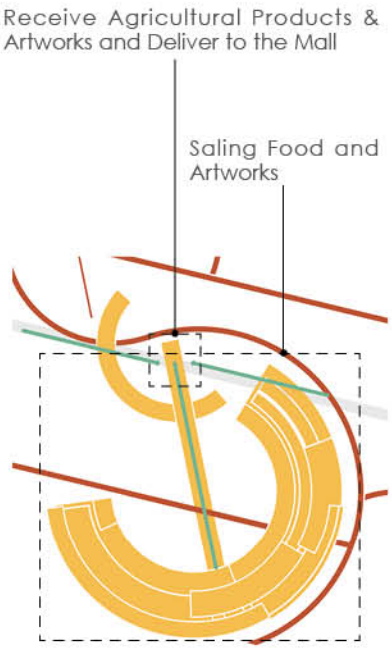
Logistics Center



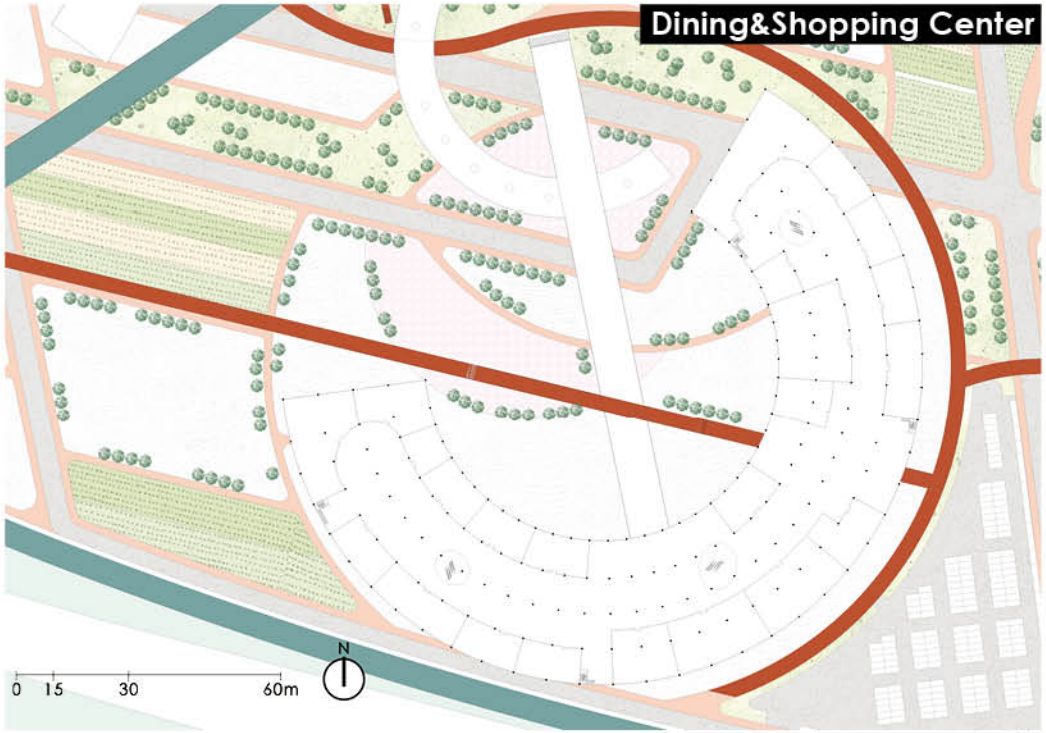






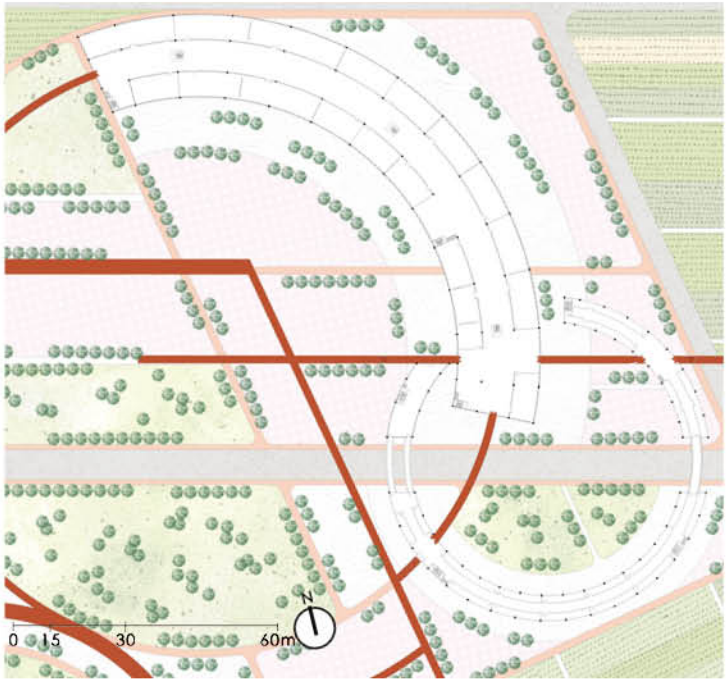


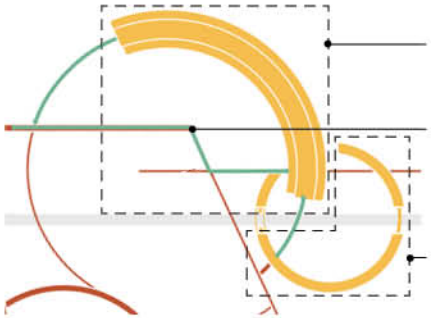
Dining&Shopping Center

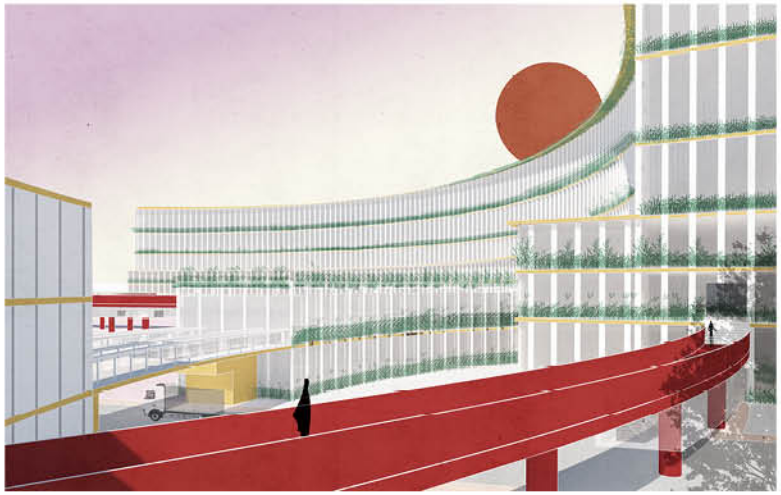


Planting Research Center

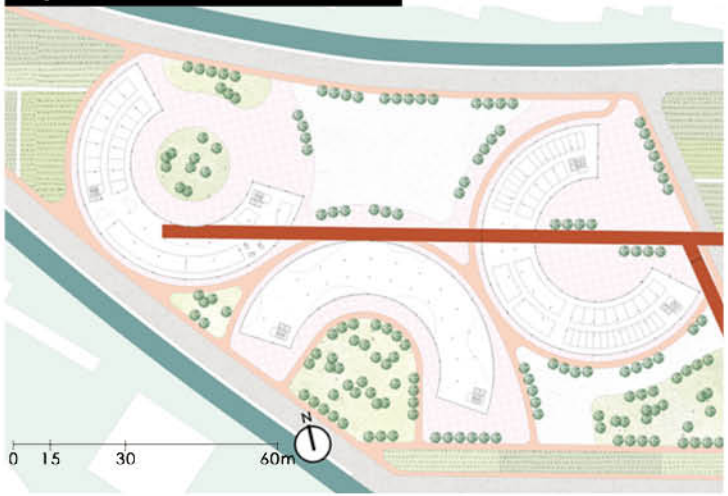








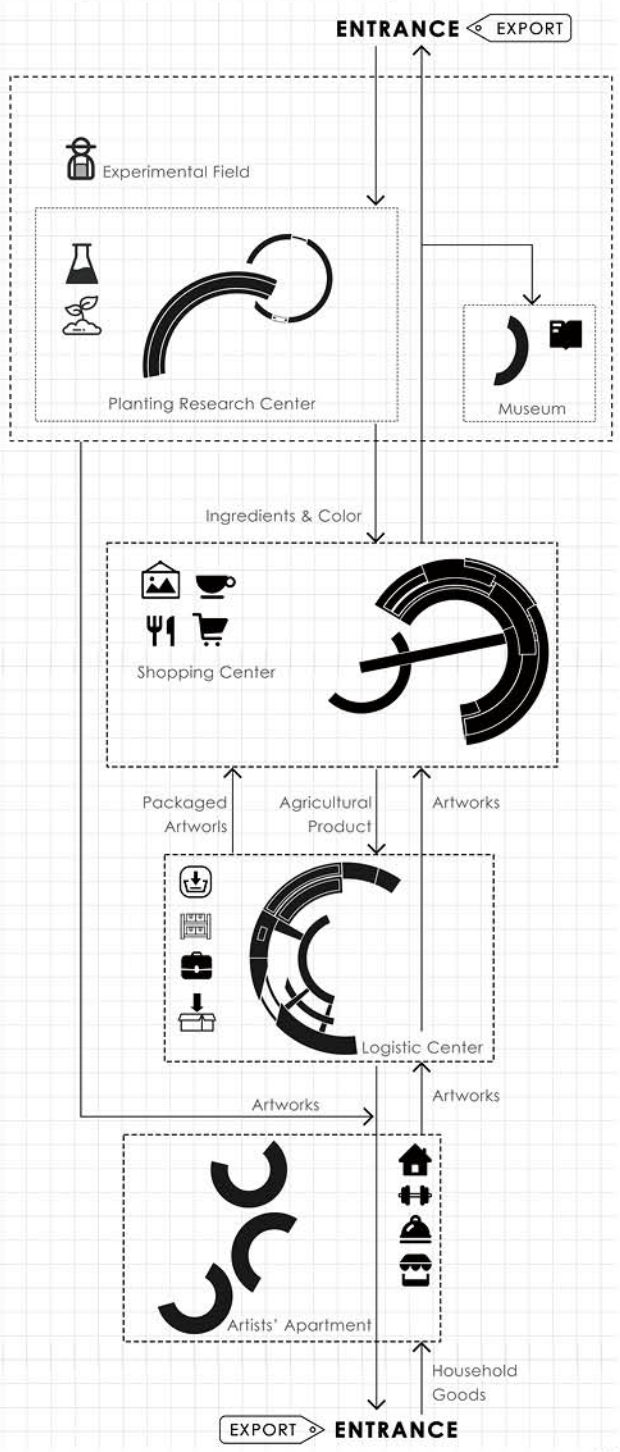
Apartment For Artists

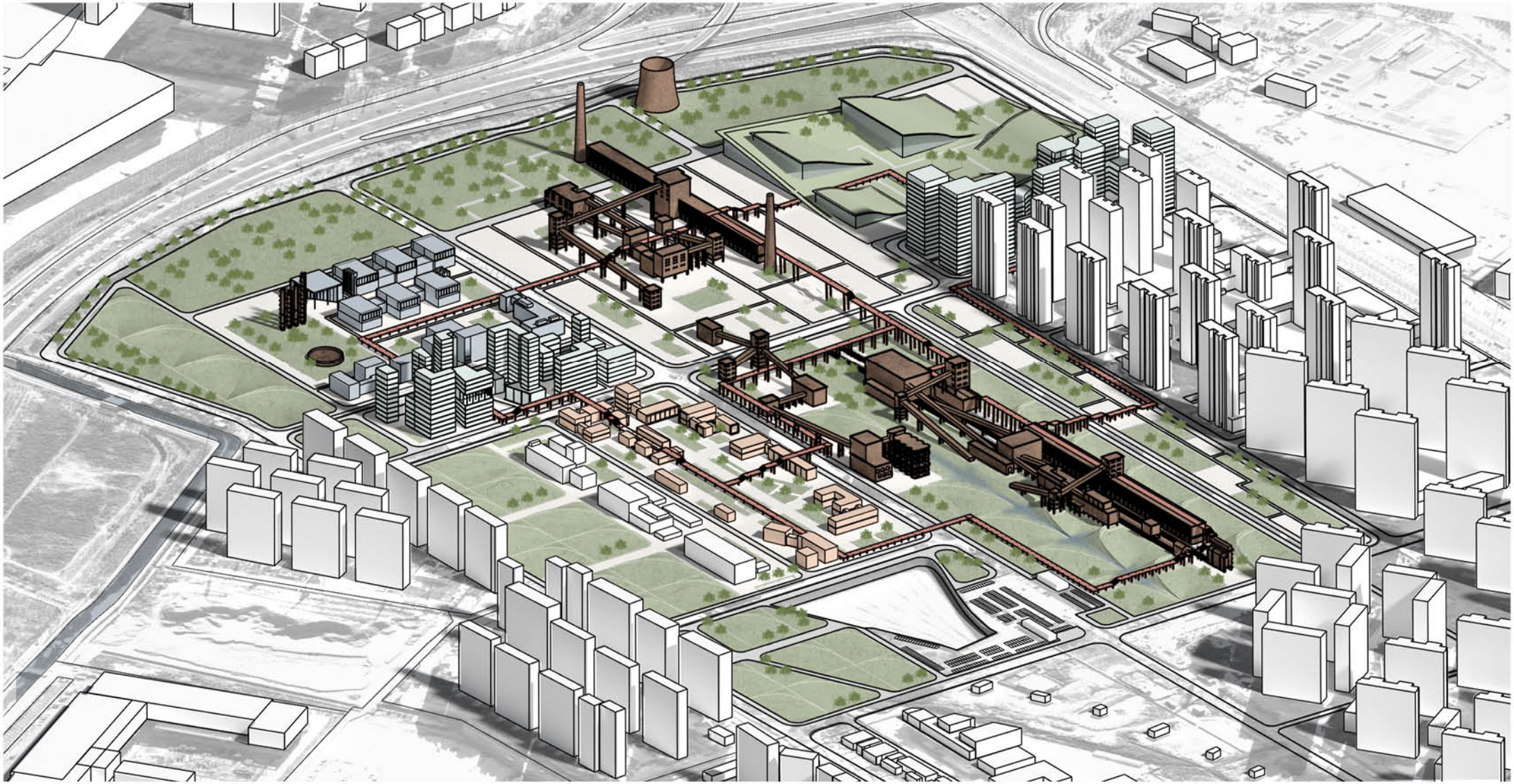






Transportation Flow Chart





Unlocking the Potential of Terrain Vague: Revitalization of the Beijing Coking Plant

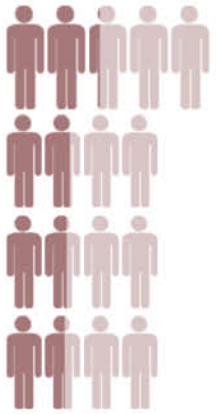
Group Work

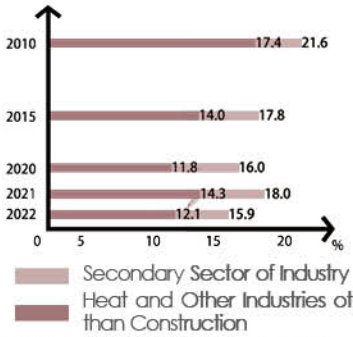
The Beijing Coking Plant, once a vital hub of the chemical industry, has become an underutilized wasteland amidst residential neighborhoods. This project seeks to reintegrate the site into the urban fabric and revitalize it as a dynamic community space. The design adopts three key strategies: replanting to restore ecological balance, reminiscence to preserve and reinterpret industrial heritage, and functional implantation to introduce contemporary uses that meet the needs of residents and visitors.

Industrial Relocation and Old Site Abandonment

Since 2010, Beijing's factories in urban areas gradually being banned or shifted to the outskirts of the city, thus prompting many factories to become empty, transforming the once-glorious center of the region into an unattended urban wasteland.

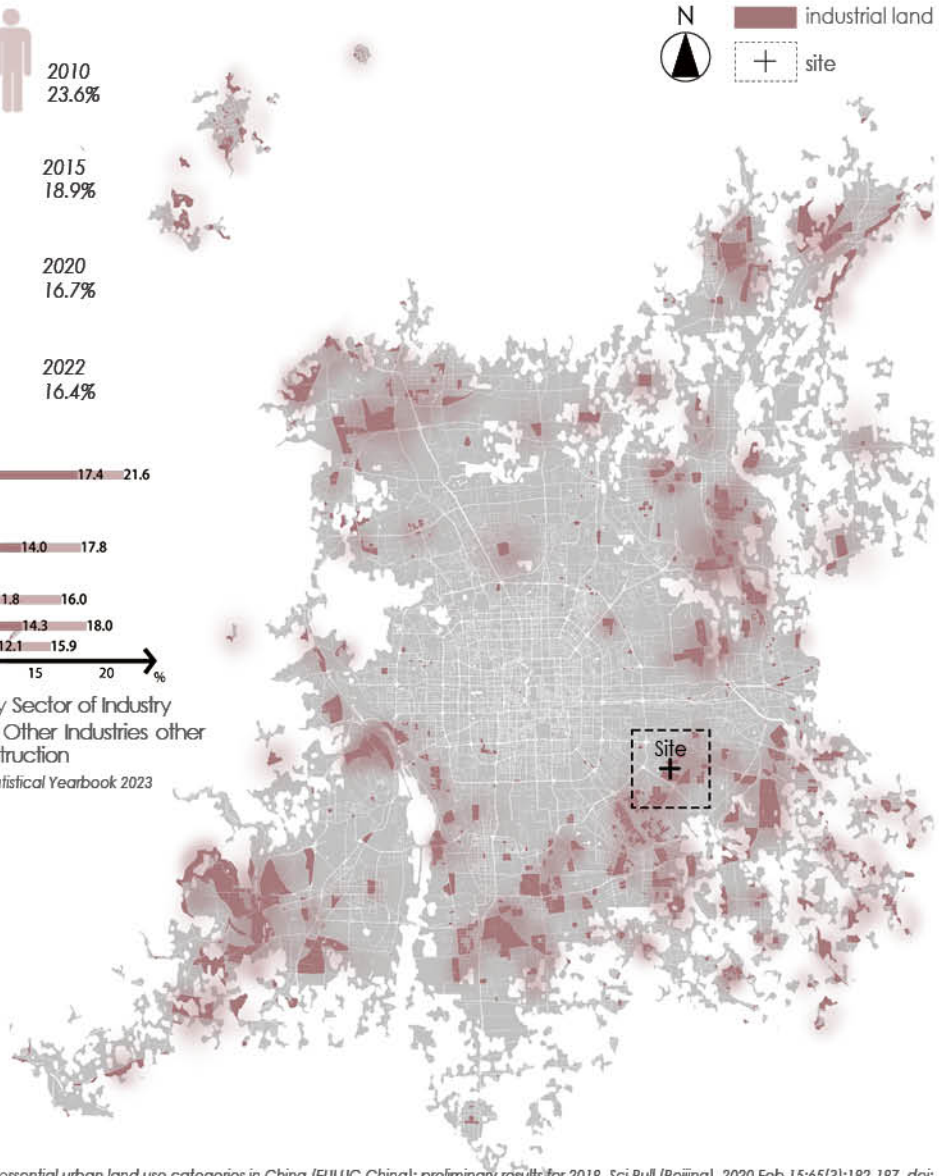
The resident population of areas employed in the coal industry is declining.





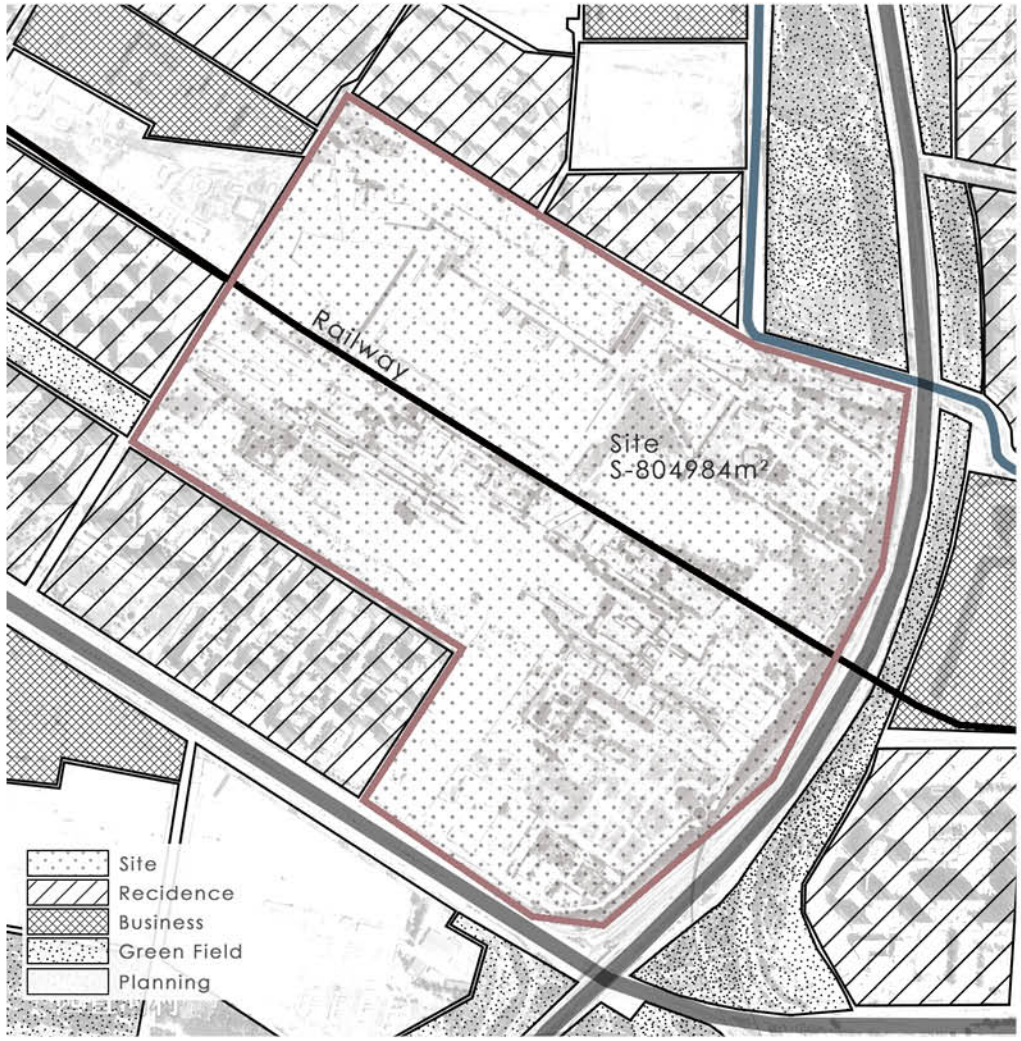
Data Reference: Beijing Statistical Yearbook 2023
Drafted By Lu

Industrial land distribution in Beijing

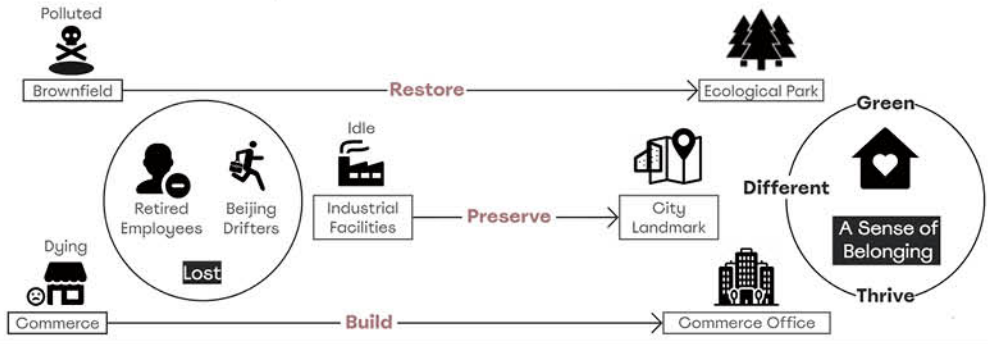


Data Reference: Mapping essential urban land use categories in China (EULUC-China): preliminary results for 2018. Sci Bull (Beijing). 2020 Feb 15;65(3):182-187. doi: 10.1016/j.scib.2019.12.007. Epub 2019 Dec 10. PMID: 36659170.
Drafted By Lu

Funtional Status of Neighboring Sites



From Lost To Belong



People Call For Change



Retired Workers

" Miss old friends

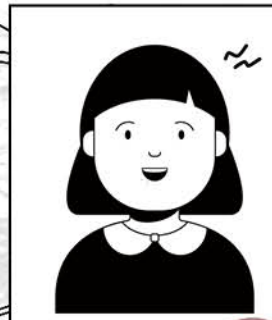
This place has been forgotten...

Too far to go to hospital

dust everywhere "

WANT—

- Re-entering the Factory
- Community Hospital
- Trees



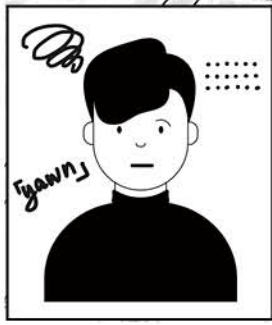
Children

" No place to play

There's nothing here. My friends won't come to play with me. "

WANT—

- More Public Space
- More Children's Recreational Facilities



Visitors/Young People

" SO BORING!!!

The industrial factory is COOOOOOOL!

Unattractive place "

WANT—

- Closer Look the Factory
- Shopping Streets
- More Interaction



Store Owners

" I'm running out of business.

No one come and buy... "

WANT—

- More Costumers

Missions From Six Aspects



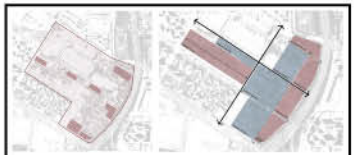
Additional infrastructure in accordance with the Beijing guidelines for complete community building services.

1. Basic Function



According to the Chaoyang Zoning Plan, a district cultural center is required to be built on the site.

2. Satisfying The Plan



Demolish abandoned buildings and transform closed, passive spaces into open activity areas.

3. Spaces



Create pedestrian corridors to connect commercial areas with residential areas and increase the vitality of commercial areas.

4. Surrounding Buildings



The original polluted bare soil will be restored through greening and planting.

5. Ground

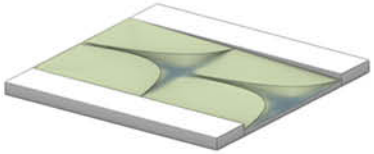


Preserve industrial facilities that characterize the site and integrate them into the site to interact with visitors.

6. Special Elements

[Strategy]

A. Restoration



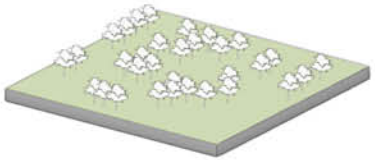
Catchment Terrain



Rainwater Garden



Vertical Greening

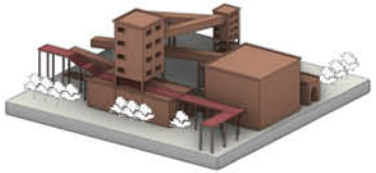


Vegetation Cover

B. Recreate



Enter the Park



Experience The Facility

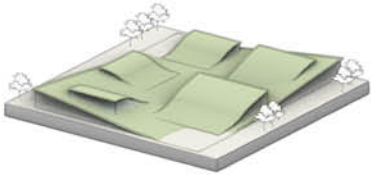


Exercise Near The Chimney



Walk Along The Path

C. Revitalization



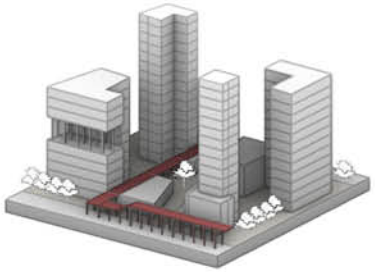
District Cultural Center



Community Service

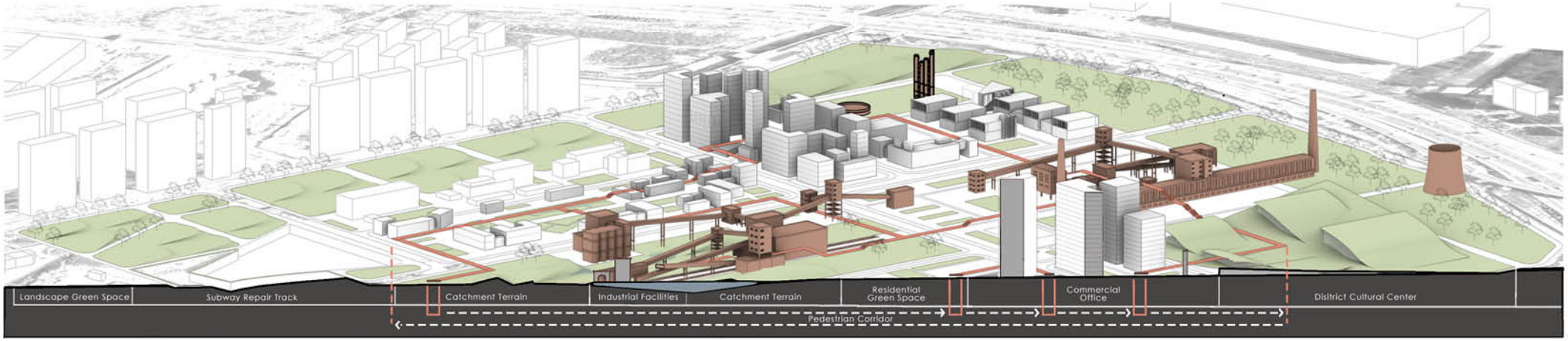


Recreational Commercial Area

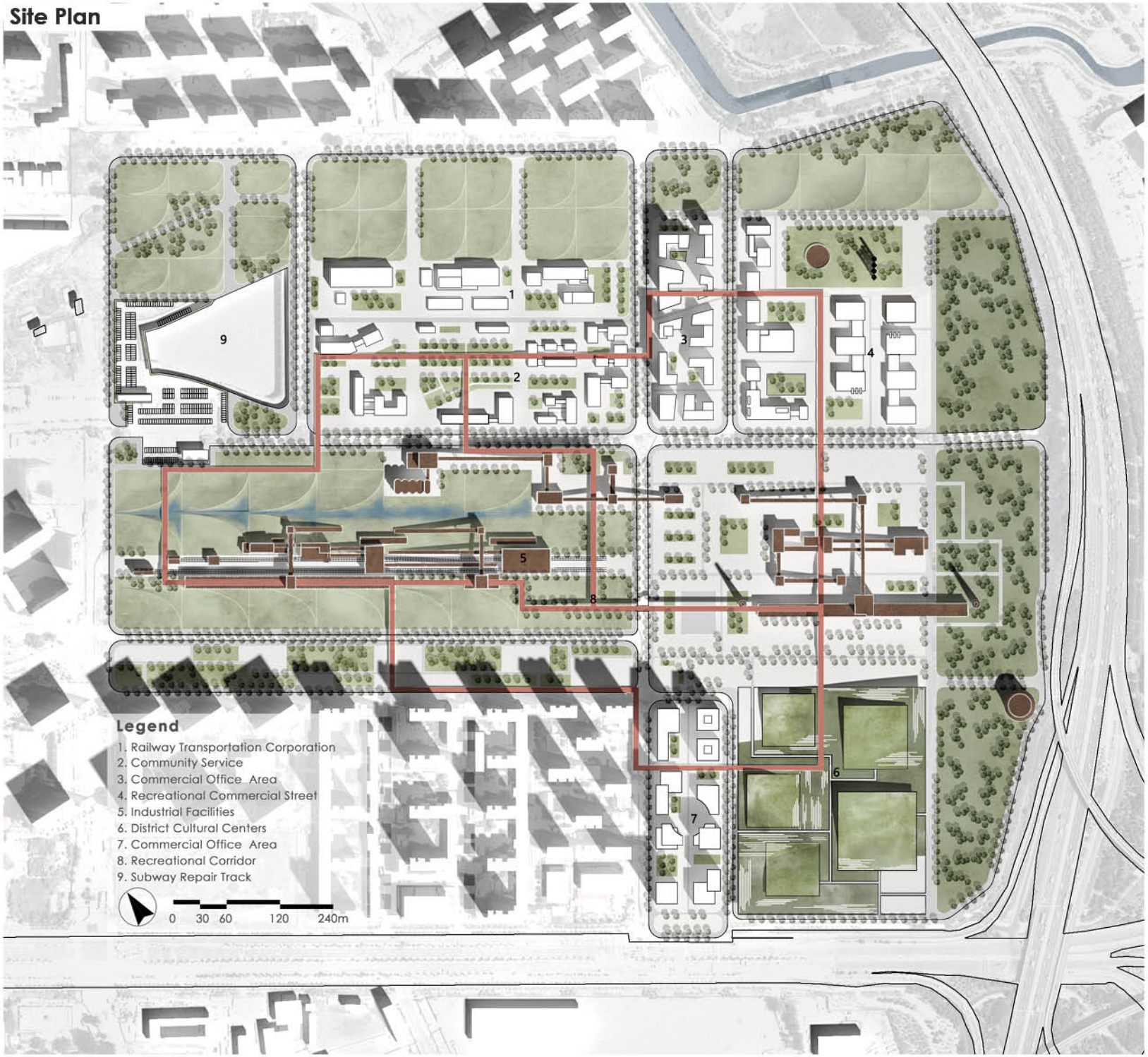


Commercial Office

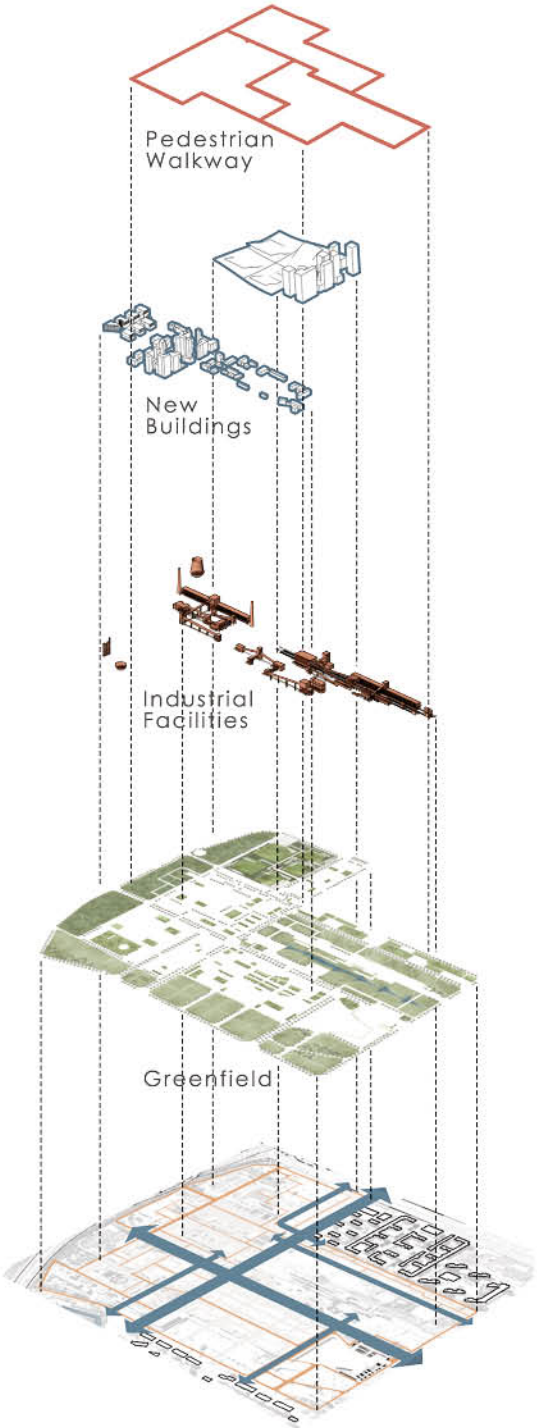
[Sectional Perspective]



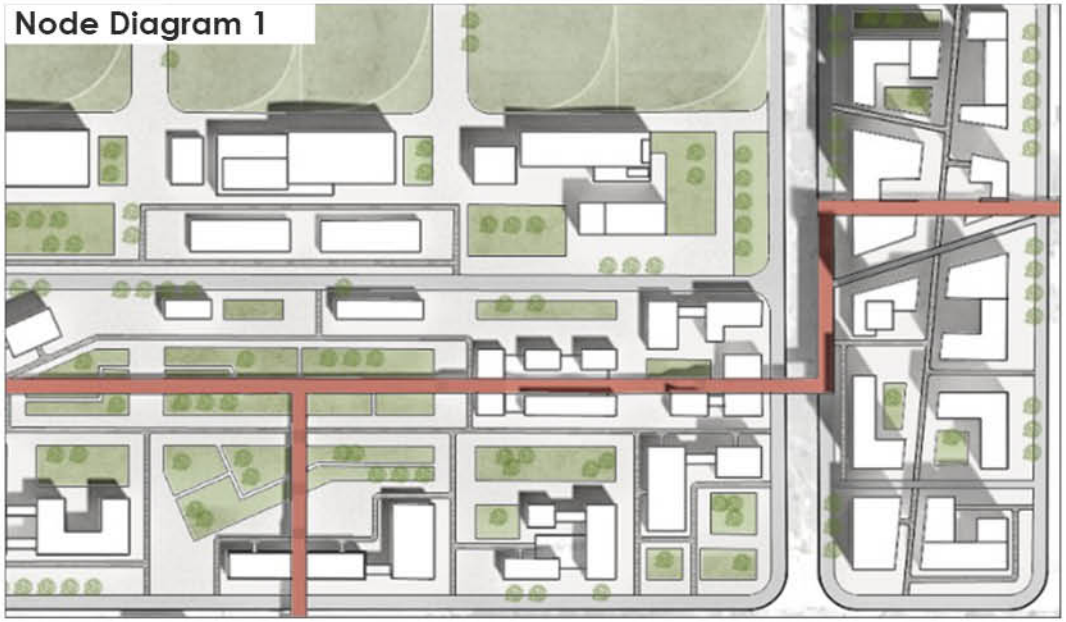
Site Plan

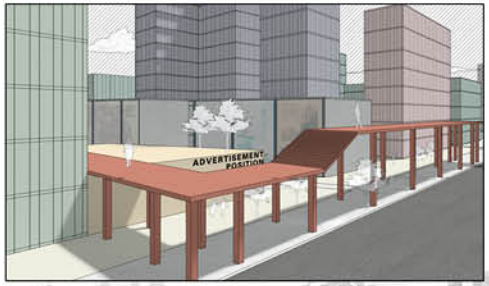


Exploded Diagram

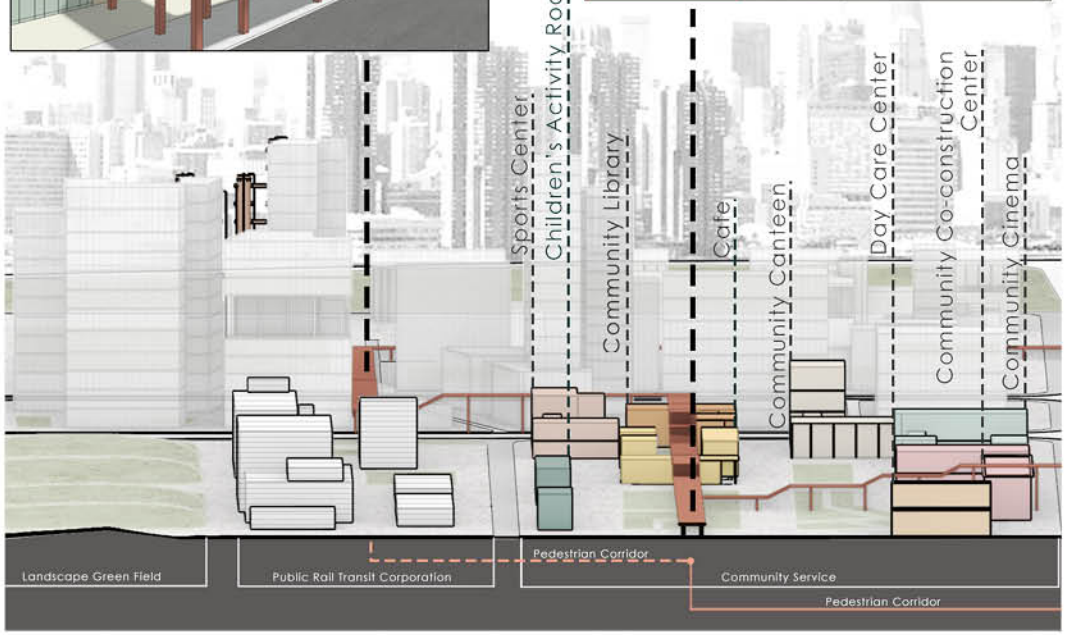


Node Diagram 1

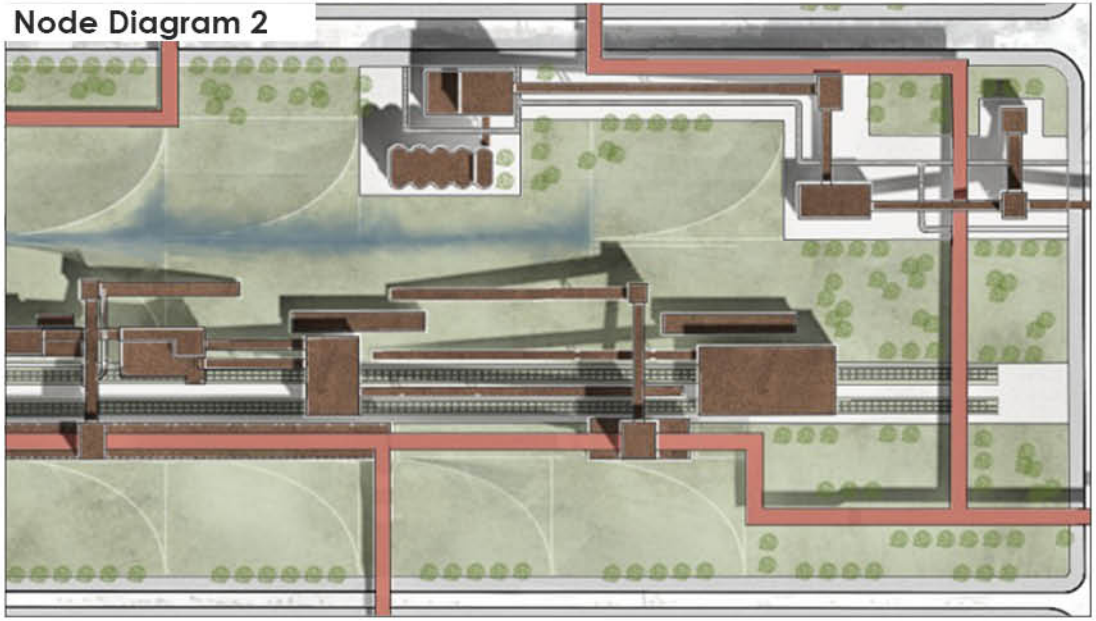


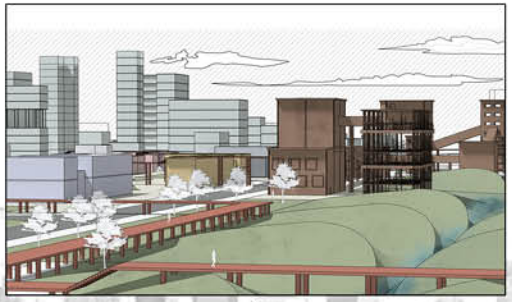


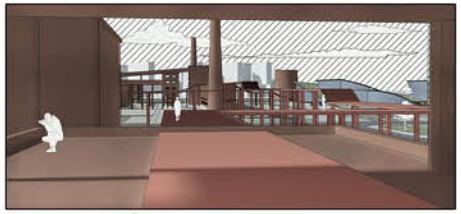


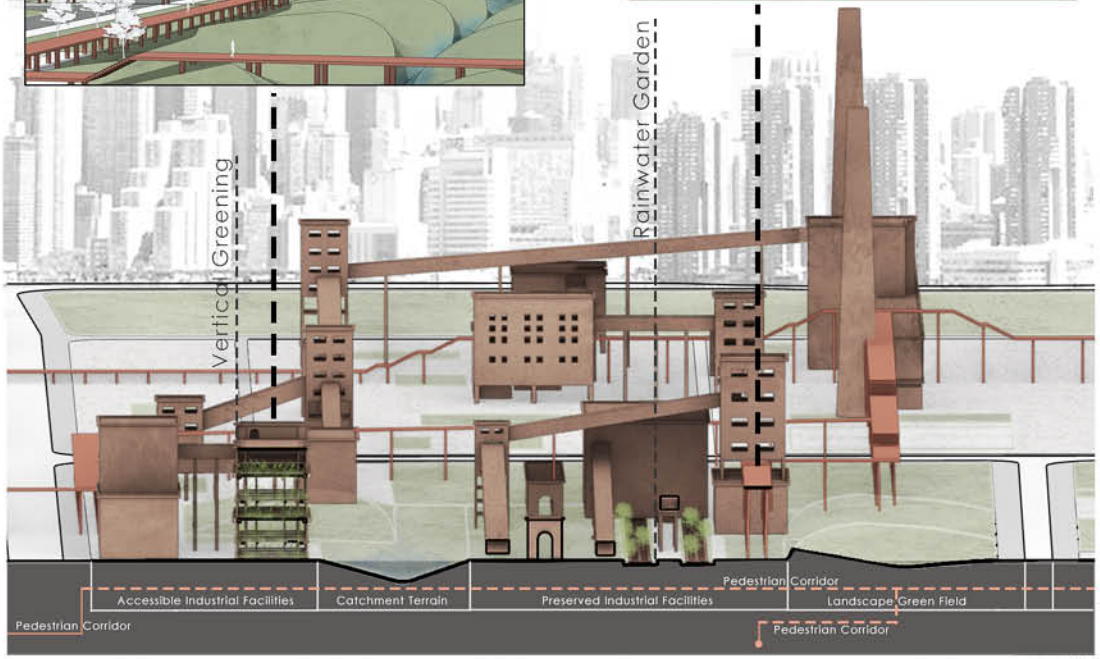


Node Diagram 2









Healing Maze

A Research on Virtual City Morphology Based on the WFC(Wave Function Collapse) Algorithm

Course Work Project: Innovation
Score: 96/100
Course Instructor: Li Ning
Project Location: Digital World
Software: Rhino, Grasshopper, Midjourney
Group/Individual: Individual
Date: 2024.4-2024.6



Concept

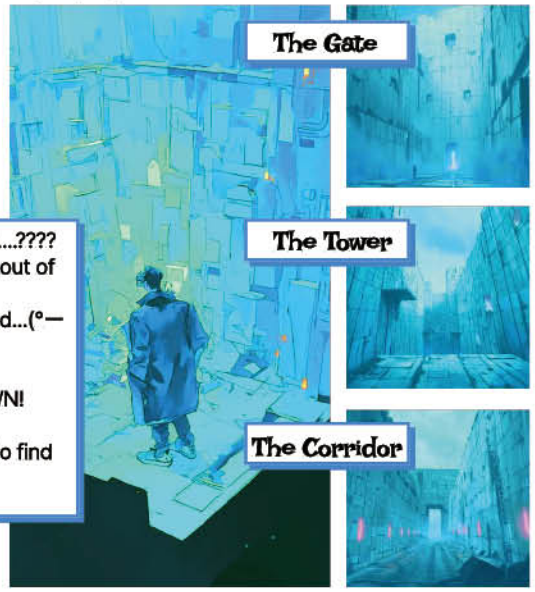
The story begins in a maze, where Old Mo explores the rules of this world, fulfills the orders of his mysterious clients, and embarks on his journey of self-discovery.

Old Mo discovered from a young age that he had "time loss syndrome," often losing consciousness for no apparent reason and waking up in a different place. Due to his poverty and the fact that it hadn't caused any harm, he never sought medical help. However, this condition led to a small mistake during his youth, which resulted in his company using it as an excuse to lay him off. In response, he decided to start a mobile restaurant, allowing him to choose his work hours freely and avoid the risk of unemployment due to any mistakes.



Generated By Midjourney

Elements



One morning, Old Mo wakes up to find himself in a completely unfamiliar place, with only his food boat by his side. He assumes that his "time loss syndrome" has acted up again and prepares to leave, only to discover that he is trapped in a maze and encounters some strange and unusual customers...

Adventures

Characters



The Main Character

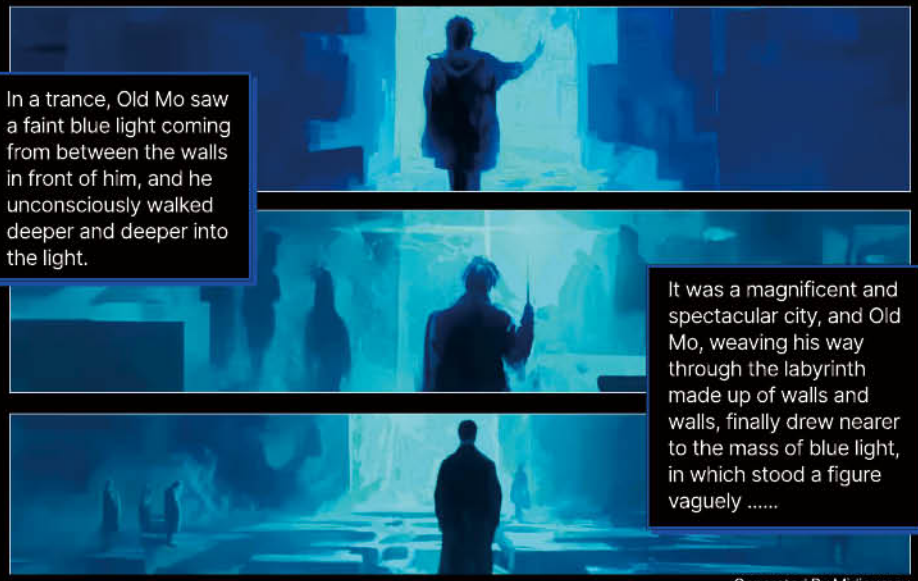
Mr. Mo

"A patient with schizophrenia, developed eight distinct personalities due to his past traumatic experiences. These personalities, both good and evil, fill the voids left by the deep wounds and regrets in his soul. However, he has forgotten his past and is unaware of the existence of these fragmented personalities within him."

Secondary Characters

- **Mysterious Clients :** These clients seem ordinary, but their orders are actually metaphors for Mo's inner world. Each order corresponds to one of Mo's split personalities.
- **The Shadow :** One of Mo's split personalities, representing the darkest part of his psyche, always trying to pull him back into his painful past.

Enter the scene



Generated By Midjourney

Mysterious Clients - Mo's Eight Split Personalities

01 Henry: Innocent Orphan

A 6-year-old boy, who was abandoned as a baby, always keens for a blueberry birthday cake. Henry makes a living by selling flowers. Mo created him at age 5 to cope with his father's abuse.

Scene Style





Cues and Tasks



02 Anne: Nurturing Grandmother

A 74-year-old kind-hearted grandmother who cares for her grandson after her husband died. She loves caramel pudding. Mo created her at age 6 after his real grandmother was killed protecting him from abuse.

Scene Style





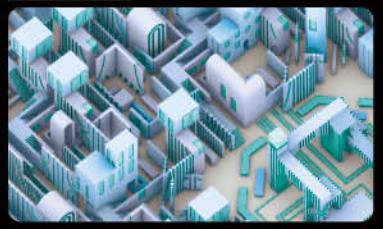
Cues and Tasks



03 Mary: Unfulfilled Dancer

A 23-year-old talented dancer who diets to maintain her figure. Her favorite food is fruit salad. Created by Mo at age 12 after his dream of becoming a dancer was crushed by his father's violence.

Scene Style





Cues and Tasks



04 Officer Hanks: Vigilant Protector

A 36-year-old strong police officer with keen observation skills, and he is fascinated with lasagna. Created by Mo at age 14 after he was disabled by an attack that the police dismissed.

Scene Style





Cues and Tasks



WHO ARE YOU NOW

05 Steve: Righteous Fighter

A 16-year-old high school student with a strong sense of justice but a violent streak. Mo created him during his high school years to protect himself from bullies. Mexican taco can calms him when angry.

Scene Style





Cues and Tasks



06 Leo: Charismatic Negotiator

A 28-year-old popular and articulate speaker, promoted during a layoff crisis. He likes Spanish paella, and is created by Mo after he was fired at age 28 due to his lack of communication skills.

Scene Style





Cues and Tasks



07 Molly: Loving Teacher

A 32-year-old elementary school teacher and Mo's wife, who loved apple pie. She was created by Mo to cope with his grief after his wife passed away because of terminal cancer.

Scene Style





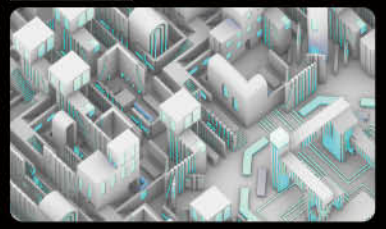
Cues and Tasks



08 Sam: Caring Doctor

A 42-year-old a veterinarian who loves Neapolitan pizza. After Mo's wife passed away, his only companion, a dog, also died. Mo created Sam's persona to cope with the loss and his unresolved feelings for his dog.

Scene Style



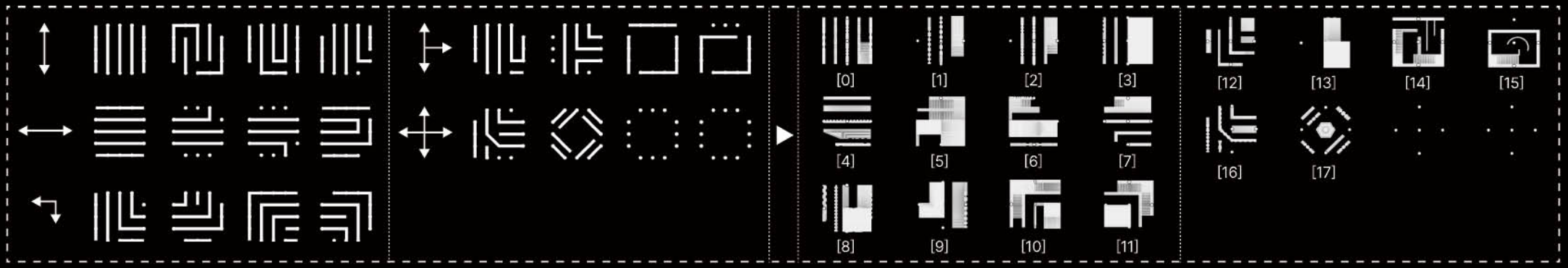


Cues and Tasks

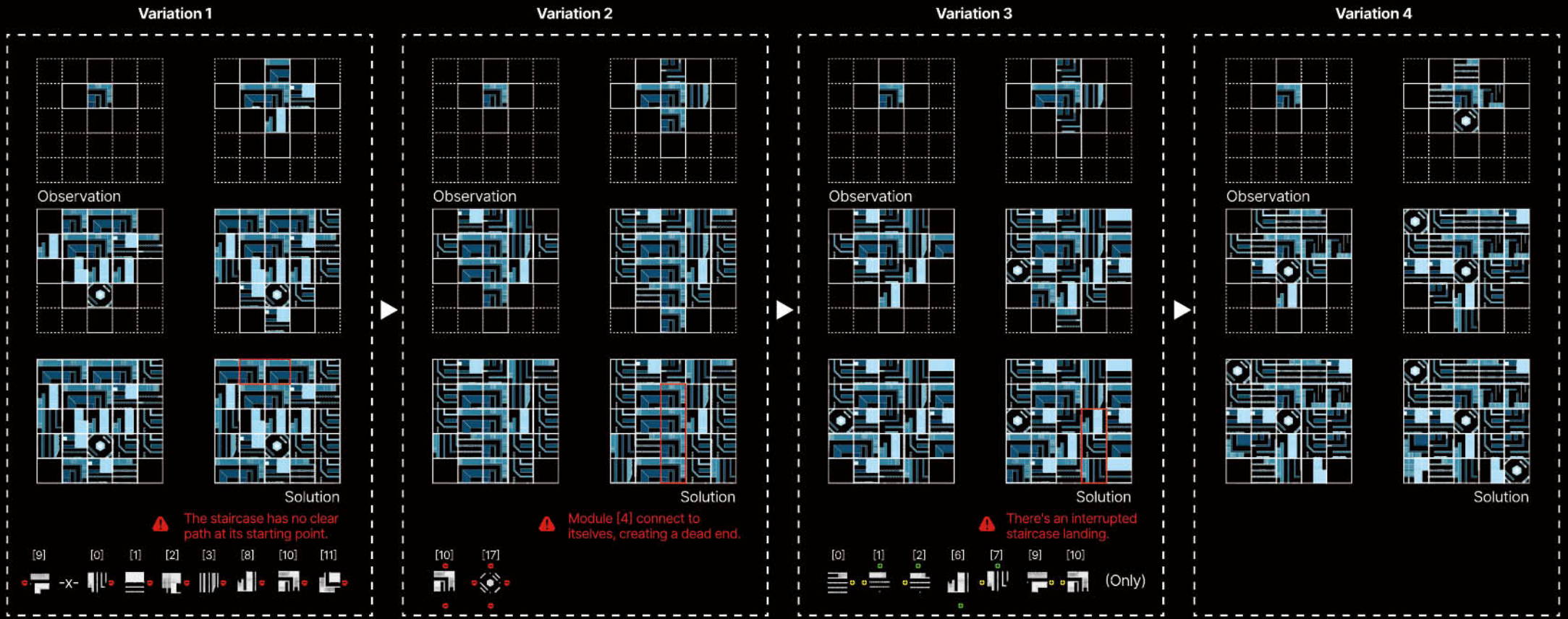


Development

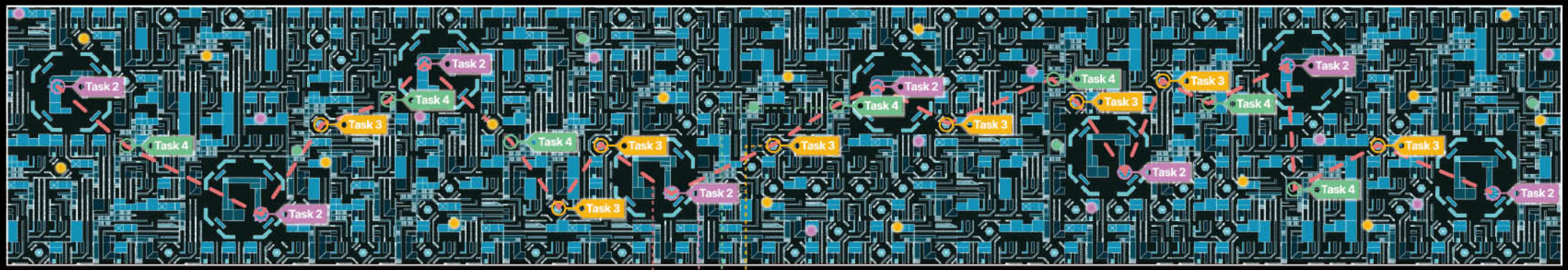
2D Modules



Rules & Generations



2D Mapping & 3D Setting

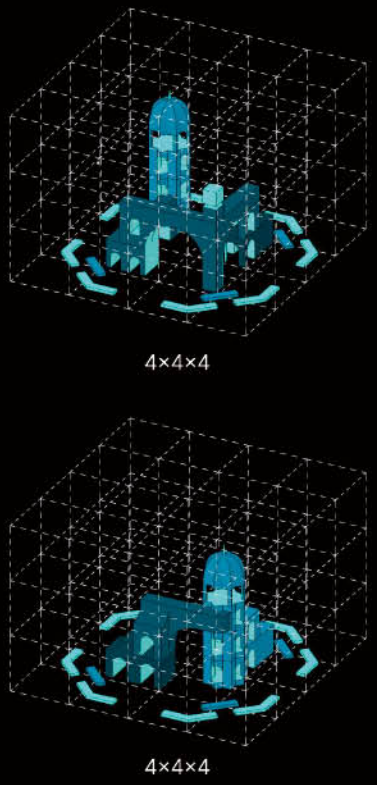


3D Modules

Basic Modules



Core Mission Nodes

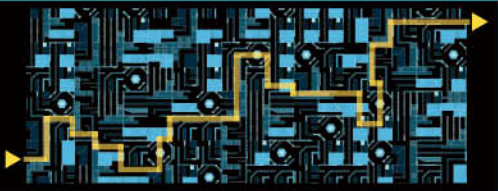


Convert 2D modules into 3D modules without altering the number of pathways in 2D modules.
Place secondary task nodes in *Basic Modules*, and add *Core Game Mission Nodes*.

Game Settings

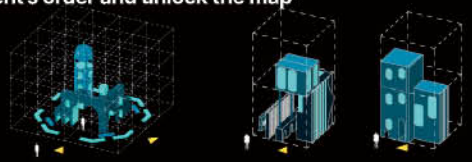
Task 1 - "Find the way out of this maze"

Walk through the labyrinth, travelling between the giant walls, discovering hidden routes and finding the EXIT.....



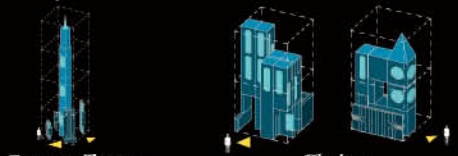
Task 2 - "Gather ingredients ot fullfill the mysterious client's order and unlock the map"

Pass through QUEST POINTS, collect the ingredients needed to fulfill the Mystery Diner's order, and solve puzzles to unlock the next level of the map!



Task 3 - "Collect treasures to gain the ability to alter the maze's paths"

Hit a dead end? Find UNUSUAL BUILDINGS, search for treasures and gain abilities to get through giant walls/alter mazes to gain access!



Task 4 - "Evade the shallow"

Beware of the consciousness guards on the way, they will prevent you from discovering the truth, and be careful to avoid the SMALL ROOMS, which are the lairs of them!



3D Game Scenes



The player - Mo, wakes up in an unknown area shrouded in thick fog. Surrounding them is a desolate landscape, with a massive maze entrance barely visible in the distance. Mo stands before the maze, his mind blank, unsure of how he arrived here, yet feeling a strange sense of familiarity. He knows that the maze ahead is somehow deeply connected to him.



As Mo enters the maze, towering walls and dark, narrow passageways surround him. Faint, ancient symbols on the walls serve as both warnings and guides. He moves cautiously, his footsteps echoing ominously, as if something could emerge from the shadows at any moment.



Deeper within the maze, Mo enters an unusual open area where objects from his past life—childhood toys, broken mirrors, old photographs—float in the air. Each item stirs distant memories, though they appear blurred and unclear. He senses that these fragments of memory might be the key to unraveling the maze's secrets.



In the maze, scattered details gradually piece together Mo's fragmented memories. He discovers rusty utensils and old photos that stir up buried pain. As the maze warps and his fractured personalities emerge, Mo realizes that fulfilling their food orders and completing hidden tasks is the key to calming their obsessions.



After completing all tasks, the final door opens to a vast, silent space with a giant mirror. Mo sees his split personalities—manifestations of his pain. He touches the mirror, the reflections blur, and the personas dissolve as the maze collapses, light guiding him to freedom.

With the maze collapsed, Mo stands in a vast open field, bathed in warm sunlight. He breathes deeply, finding inner peace, ready to embrace a new beginning despite the uncertainties ahead.



Historical and Cultural Elements Research and Enhancement from a Pedesetrian Perspective

Course Work Project: Urban and Rural Infrastructure Planning
Course Instructor: Yu Chaingming
Project Location: Beijing, Dongcheng District
Software: Sketchup, Arcgis, Photoshop
Group/Individual: Group(Lu Yutian, Quan Zhiyi)
Contribution: main concept: 50% ; arcgis: 70% ; drawings: 100%; Design: 90%; Stastics: 20%
Date: 2023.9-2024.1

Based on the pedestrian perspective, this project investigates the quality and quantity of heritage sites in Nanluoguxiang through field research and interviews, explores the current status of heritage sites in Nanluoguxiang, and designs and builds facilities such as heritage kiosks according to the needs of the population and the basic conditions of the hutongs, so as to explore the historical and cultural heritage of Nanluoguxiang and to increase the fun of walking.

2. Status Analysis

2.1 Functional Land Use Analysis



Distribution of Land Uses in the Nanluoguxiang Area

The land within the study area is predominantly residential, with commercial areas distributed along both sides of Nanluoguxiang Lane. Two underground station entrances are located to the west and south, and a parking facility is situated to the south. Municipal facilities and multifunctional service areas are evenly distributed within the residential zone to meet community needs. Notably, there are no green spaces or public squares within the area.

2.2 Characteristics of Street Space Use and Key Building Features

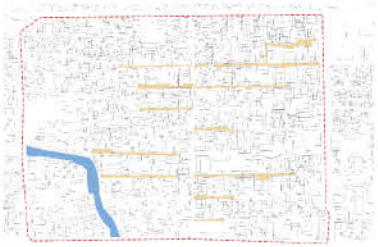


The analysis identifies the actual uses of the street space and compares the distribution of commercial and historical-cultural spaces.

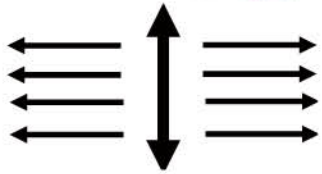
2.2.1 Commercial Spaces

Commercial activity is primarily concentrated on the north side of Gulou East Street and along both sides of the north-south axis of Nanluoguxiang. There is a notable absence of significant commercial spaces in the east-west alleys.

2.2.2 Historical and Cultural Spaces



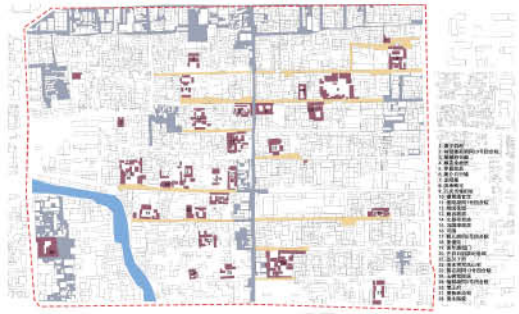
Historical and cultural spaces are predominantly distributed along the east-west hutongs, adjacent to residential buildings.



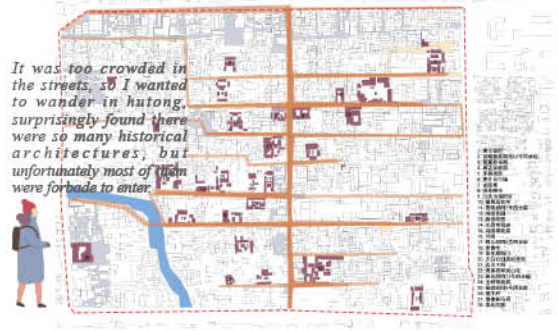
The street layout of Nanluoguxiang follows a fishbone structure, with commercial spaces forming the main axis and historical-cultural spaces branching off, creating clear distinctions and a unique distribution pattern.

2.2.3 Distribution and Characteristics of Key Buildings

The buildings are mainly former residences of notable individuals, densely distributed with abundant historical and cultural resources. These buildings are primarily located in the east-west hutongs, some distance from the commercial spaces along the north-south axis of Nanluoguxiang.

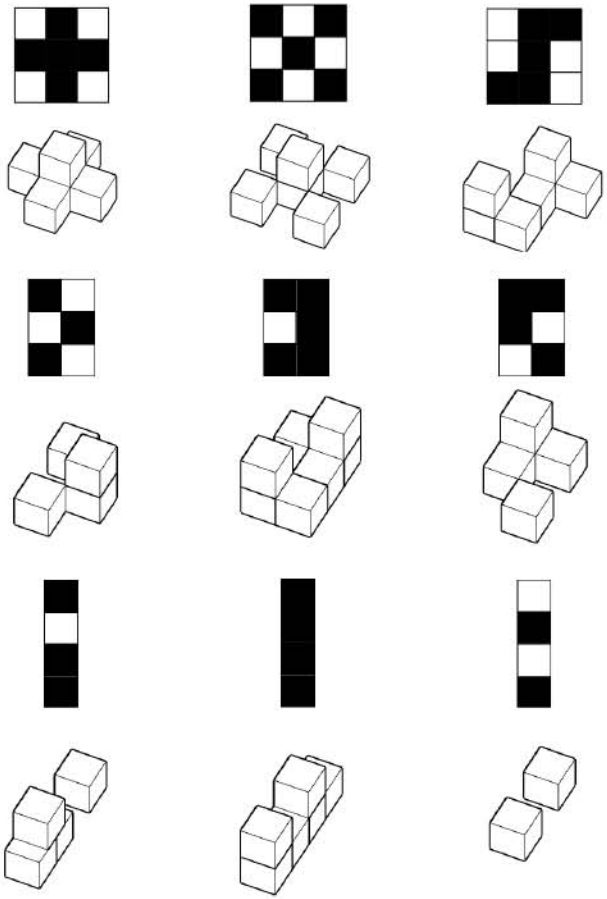


During the research, twelve tourists were randomly invited to trace their preferred walking routes on a floor plan. The analysis of these routes revealed that tourists generally preferred walking along the north-south axis of Nanluoguxiang, focusing their activities in the commercial areas. They occasionally ventured into the east-west hutongs but typically returned to the main commercial axis upon discovering that the historical buildings were not accessible to the public.



4.4 Module Combinations

The design modules are flexible and can be adapted according to the social needs of the users. During the design process, the unused spaces in the hutongs were categorized into three types: square, rectangular, and strip. Possible combinations of these modules were explored to maximize their functionality.



4.5 Strategies

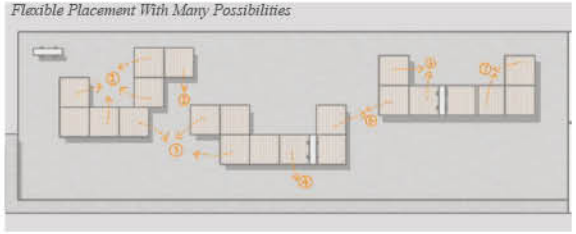
4.5.1 Flexible Space

The space in the hutongs is compact, yet there are still areas where people can rest. Given the modular and movable nature of the design, the modules can be stacked in the corners when not in use, ensuring that they do not obstruct the flow of the hutong. However, when needed, the modules provide spaces for both residents and tourists to rest.

4.5.2 Removeable Modules

Removable modules offer flexibility for different user groups, allowing individuals or groups to decide whether they wish to sit alone or together, socialize, play chess, or enjoy a snack. The design aims to create a variety of interactive spaces that encourage spontaneous human interaction.

Flexible Placement With Many Possibilities



4.5.3 Multiple Uses for Signage

■ Introduction to Historical and Cultural Buildings

To address the challenge of tourists not being able to enter private properties, a brief exhibition of 2-3 historical buildings in the neighborhood will be held in the unused space. This will be achieved through signage featuring introductory text engraved in frosted glass, with a miniature model of the building placed in the center.





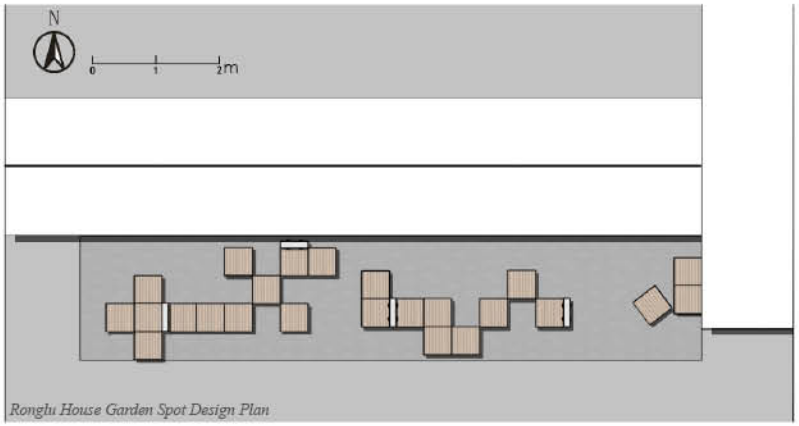
■ Night Lighting

Research revealed that some hutongs lack street lighting, contributing to low security at night. The spotlights in the signage can also serve as functional lighting for these areas.

■ Non-Movable Modules

To prevent the encroachment of motorized vehicles and ensure that the spaces are preserved for pedestrian use, non-movable modules will be incorporated into the design to prevent parking in these areas.

4.6 Master plan



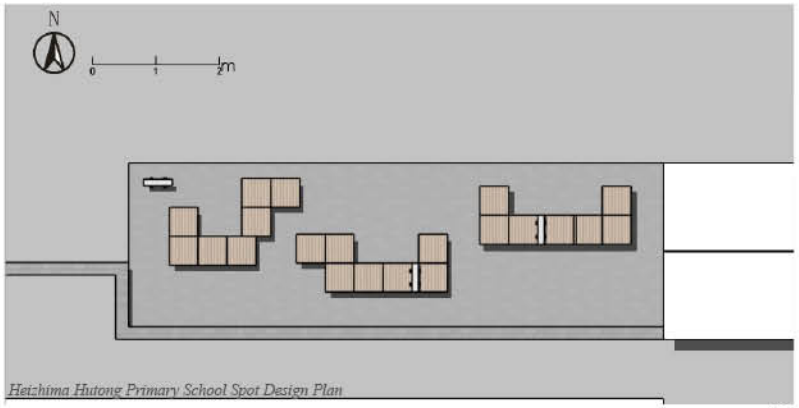
Ronglu House Garden Spot Design Plan











Heizhima Hutong Primary School Spot Design Plan